

Towards DQI for finance applications

Victor Onofre¹, Maria E. S. Castro¹, Anni Korin², and Joni Ranta²

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Decoded Quantum Interferometry (DQI) is a new full quantum (not hybrid) optimization algorithm whose advantage hinges on a single requirement: the problem must represent the parity-check matrix of a error-correcting code. For generic problems this structure is absent. We identify fixed-income immunization as a finance problem that natively fulfills the structure. The fundamental goal is to ensure that a financial institution can always meet its future payment obligations (liabilities), regardless of whether interest rates go up or down. We formalize a finite-field, combinatorial surrogate of immunization, cast it as a max-XORSAT instance, and show via statevector simulation that DQI amplifies good solutions only when this algebraic code structure is present, at low circuit cost. We further implement small toy instances in the 53-qubit superconducting processor of IQM, observing above-random amplification. These results do not constitute a production immunization optimizer, rather, they establish the problem as a finance-motivated structure in which DQI's decoding requirement is naturally satisfied.

1 Introduction

Decoded Quantum Interferometry (DQI) is a quantum optimization algorithm introduced by Jordan, Shutter, Wootters, Zalcman, and collaborators (2024) [2]. It targets the max-LINSAT

Victor Onofre: vonofre68@gmail.com

Maria E. S. Castro: maria.silvacastro@aalto.fi

Anni Korin: anni.korin@aalto.fi

Joni Ranta: joni.ranta@aalto.fi

and its binary special case max-XORSAT: given a system of linear constraints $B \cdot x = v$ over a finite field, find the assignment x that satisfies as many constraints as possible. DQI's headline result is that for some structured instances, it produces approximate solutions that beat the best known classical algorithms, making it one of the more credible candidates for a genuine quantum optimization advantage.

The central idea is a reduction from optimization to decoding. DQI prepares a quantum superposition over candidate solutions, weighted by a carefully chosen polynomial in the number of satisfied constraints, and then applies a quantum Fourier transform. After this transform, the objective "how many constraints does x satisfy" is re-expressed in a dual basis where it looks exactly like an error-correcting-code decoding problem: the amplitude on a state corresponds to how well a received word can be decoded against the code whose parity-check matrix is B . DQI doesn't search the solution space directly; it interferes with amplitudes so that the optimization answer falls out of decoding a code. The quality of the final solution is governed by how good a decoder you can run.

The decoding step is only feasible to implement when B is a parity-check matrix with structure. When B is the parity check of an algebraically decodable code (BCH or Reed–Solomon), the decoding step is the Berlekamp–Massey algorithm, which has a structured, reversible circuit. So DQI amplifies precisely on instances whose constraint matrix is algebraic.

DQI thus turns optimization into decoding. The open problem is finding real problems whose structure makes that decoding cheap. In this work we argue that *fixed-income immunization* has exactly such structure. A bond portfolio's interest-rate sensitivities (duration, convexity, higher key-rate durations) are successive moments of maturity, so "match a liability's sen-

sitivities” becomes a Vandermonde/locator system, precisely the parity-check structure of a BCH/Reed–Solomon code. A genuinely financial question, “which few bonds break the match between portfolio and liability?”, then maps onto the same algebraic decoding DQI already runs efficiently, rather than onto the expensive belief-propagation circuit forced on a generic constraint matrix.

Scope and contribution. We are deliberate about what is and is not claimed. We do *not* claim to solve the real/industrial immunization problem. Our contribution is threefold:

1. **Structural match.** Immunization’s moment-matching constraints are Vandermonde matrices, structurally identical to BCH/RS parity checks, exactly the algebraically decodable structure DQI amplifies cheaply.
2. **A finite-field surrogate that DQI can run.** We define a *combinatorial*, finite-field surrogate of immunization (Sec. 4), “which subset of bonds is the odd-one-out for the binarised moment syndrome?”, and pose it as the max-XORSAT instance DQI solves. The mapping $\mathbb{R} \rightarrow \text{GF}(2^m)$ is a modelling choice, made explicit below. What is preserved is the algebra, not the real-valued cash flows.
3. **Simulation and hardware evidence.** Statevector simulations for amplification to the algebraic-decoder regime, and we run toy instances on the VTT Q50 processor.

2 DQI

Section based on [2, 4]

DQI is an algorithm for the max-LINSAT optimization problem. One is given n variables $\vec{x}(x_1, \dots, x_n) \in \mathbb{F}_p^n$ over a finite (Galois) field, and m linear constraints

$$\vec{b}_i \cdot \vec{x} = \sum_j B_{ij} x_j \in T_i, \quad i \in [1, m] \quad (1)$$

where B is an $m \times n$ matrix over $\mathbb{F}_p$ and each $T_i \subseteq \mathbb{F}_p$ (with $|T_i| = r$) is a target set of allowed values. The goal is to satisfy as many of the m constraints as possible. This is encoded as an

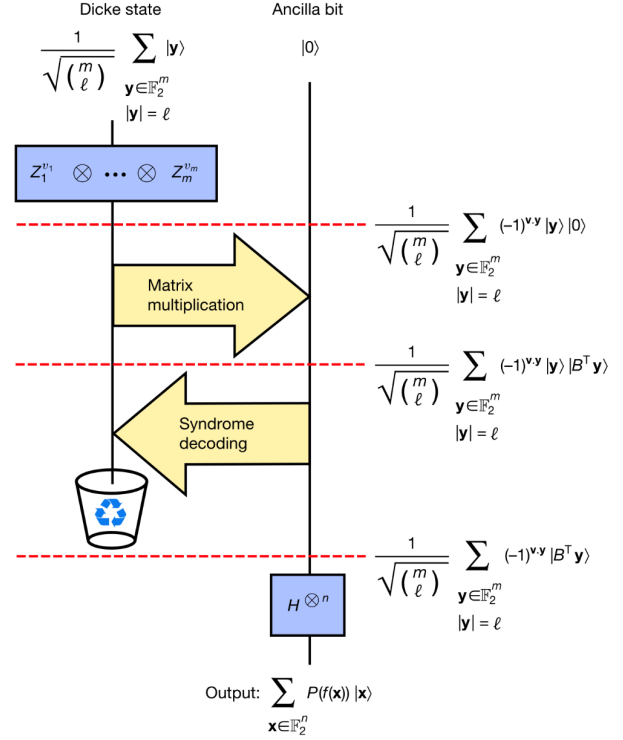


Figure 1: DQI algorithm

objective function by defining $f_i(y) = +1$ if $y \in T_i$ and -1 otherwise, so that

$$f(\vec{x}) = \sum_{i=1}^m f_i(\vec{b}_i \cdot \vec{x}) = \#\text{SAT} - \#\text{UNSAT} \quad (2)$$

$$\text{maximize } f(\vec{x}) \quad (3)$$

Special cases include max-XORSAT (binary, $p = 2$, $r = 1$, with $f_i(y) = (-1)^{y+v_i}$) and the Optimal Polynomial Intersection (OPI) problem, where B is a Vandermonde matrix $B_{ij} = y_i^j$ and one seeks the degree- $(n-1)$ polynomial passing through the most target sets. These problems are NP-hard in the worst case, which is what makes them interesting targets for a quantum speedup.

The central idea is not to search, but to prepare a quantum state whose amplitudes encode the objective function, so that measuring it preferentially yields high-scoring $\vec{x}$. A naive version would prepare $\propto \sum_{\vec{x}} (f(\vec{x}) + m) |\vec{x}\rangle$ (the $+m$ keeps amplitudes non-negative since $f \in [-m, m]$). DQI improves on this by applying a degree- ℓ polynomial to the objective,

$$P(f(\vec{x})) = \sum_{k=0}^{\ell} \alpha_k f^k(\vec{x}) \quad (4)$$

$$|\text{DQI}\rangle = |P(f)\rangle := \sum_{\vec{x} \in \mathbb{F}_p^n} P(f(\vec{x})) |\vec{x}\rangle. \quad (5)$$

Because $(f(x) + m)^{2\ell} \geq (f(x) + m)^2$, raising the objective to a power sharpens the distribution toward the best solutions; the coefficients α_k (equivalently the normalized weights w_k) are tuning parameters that control how much the good solutions are amplified.

The crucial manipulation is to rewrite this state in a form that a quantum computer can build. Since $f = \sum_{i=1}^m f_i$ is a sum of constraint functions and each satisfies $f_i^2 = 1$ (they are ± 1), powers of f collapse onto elementary symmetric polynomials $P^{(k)}(f_1, \dots, f_m)$, unweighted sums of products of k distinct constraints:

$$P(f) = \sum_{k=0}^{\ell} u_k P^{(k)}(f_1, \dots, f_m), \quad (6)$$

$$|\text{DQI}\rangle = \sum_{\vec{x}} \sum_{k=0}^{\ell} u_k P^{(k)}(f_1, \dots, f_m), |\vec{x}\rangle. \quad (7)$$

Each product $f_{i_1} f_{i_2} \dots f_{i_k}$ encodes the joint satisfaction parity of k constraints, and the constructive interference between these terms is precisely what amplifies good solutions, hence "interferometry." For max-XORSAT this can be made fully explicit writing $f_i = (-1)^{(B^T \vec{y}) \cdot \vec{x} + v_i}$ and restricting to vectors $\vec{y} \in \mathbb{F}_2^m$ of Hamming weight $|\vec{y}| = k$, the state becomes

$$|\text{DQI}\rangle = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{\ell} \frac{w_k}{\sqrt{\binom{m}{k}}} \sum_{|\vec{y}|=k} \sum_{\vec{x}} (-1)^{\vec{y} \cdot \vec{y} + (B^T \vec{y}) \cdot \vec{x}} |\vec{x}\rangle.$$

The vector $\vec{y}$ here is a weight- k error pattern, and $B^T \vec{y}$ is exactly a syndrome. This is the coding-theory connection: amplifying the objective is mathematically the same as working with the dual code of B .

The algorithm prepares this state using three registers: weight (W), error (E), and syndrome (S), in seven unitary steps:

1. Prepare weights: build $|\psi_1\rangle = \sum_{k=0}^{\ell} w_k |k\rangle_W$ in the W register ($\lceil \log_2(\ell+1) \rceil$ qubits).
2. Prepare Dicke states: conditioned on k , create the Dicke state $|D_k^m\rangle = \binom{m}{k}^{-1/2} \sum_{|\vec{y}|=k} |\vec{y}\rangle$ in the E register. The uniform superposition over weight- k error patterns.

3. Disentangle the weight register, uncomputing W so the state lives entirely in E : $\sum_k \frac{w_k}{\sqrt{\binom{m}{k}}} \sum_{|\vec{y}|=k} |\vec{y}\rangle_E$.
4. Add constraint phases: apply $U_4 = Z^{v_1} \otimes \dots \otimes Z^{v_m}$ to imprint the $(-1)^{\vec{v} \cdot \vec{y}}$ factor.
5. Matrix multiplication: computes $|B^T \vec{y}\rangle$ into the S register.
6. Disentangle the error register by applying $D_B(B^T \vec{y}) = \vec{y}$
7. Fourier transform: apply $H^{\otimes n}$ (for $p = 2$), which converts $|B^T \vec{y}\rangle$ into the phase $(-1)^{(B^T \vec{y}) \cdot \vec{x}}$, yielding $|\psi_7\rangle = |\text{DQI}\rangle$.

For general max-LINSAT the structure is identical. The differences are that Step 4 imprints the Fourier-transformed constraint functions $\tilde{g}_i(y_i)$ (built from the shifted/rescaled $g_i = (f_i - f)/\varphi$), and Step 7 uses the full quantum Fourier transform $U_7 = F^{\otimes n}$ instead of Hadamards. Measuring the S register then returns a high-scoring $\vec{x}$.

The single non-trivial step is Step 6. This is exactly a decoding problem: B plays the role of a linear code's parity-check/generator matrix, and DQI works efficiently only when that code admits an efficient classical decoder. DQI thus out-sources the hard combinatorial work to a classical decoder. The QFT of a superposition over a code C produces a superposition over its dual $C^\perp$, mapping the optimization objective onto syndromes.

The payoff is quantified by the semicircle law for the optimal weights:

$$\frac{\langle s \rangle}{m} = \left(\sqrt{\frac{\ell}{m} \left(1 - \frac{r}{p}\right)} + \sqrt{\frac{r}{p} \left(1 - \frac{\ell}{m}\right)} \right)^2.$$

The claim is that matching DQI's $\langle s \rangle$ classically would require superpolynomial time.

3 Fixed-income immunization

Section based on [1]

Pension funds, insurers, and banks live with a structural mismatch: they owe a stream of liabilities (annuities, pension payments, etc) that fall due far in the future, and they fund those promises with a portfolio of bonds (the assets).

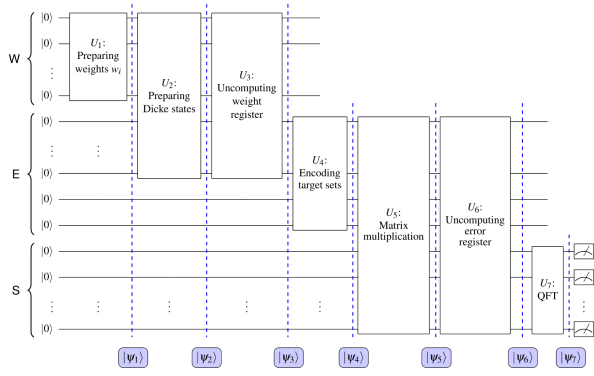


Figure 2: DQI Circuit

Both sides of the balance sheet are just discounted cash flows, so both move when interest rates move. The danger is that they move by different amounts. When that happens, equity, assets, and liabilities swing, and a large enough adverse swing is an insolvency event. This is not abstract: the 2023 collapses of Silicon Valley Bank and First Republic were, at root, a failure to hedge long-dated liabilities against a rate move.

The present value of a liability paying f_n at time t_n under a continuously-compounded rate r is

$$P = \sum_{n=1}^N e^{-rt_n} f_n.$$

Each term shrinks as r rises, so P falls when rates rise and the asset side does too. The hedging question is: how do we build a bond portfolio whose value tracks the liability's value as rates move, so equity stays put?

The answer is to match sensitivities, not just values. The first-order sensitivity of present value to a rate change is the duration,

$$D = -\frac{\partial \log P}{\partial r} = \frac{1}{P} \sum_{n=1}^N t_n e^{-rt_n} f_n = \sum_{n=1}^N w_n t_n,$$

$$w_n = \frac{e^{-rt_n} f_n}{P},$$

which is just the value-weighted average time to payment. If the portfolio's duration equals the liability's duration, then to first order a small parallel shift in rates changes assets and liabilities by the same amount. Matching the second-order term as well (convexity) protects against larger moves, and matching local sensitivities at

specific maturities (key-rate durations) protects against the curve twisting rather than shifting in parallel.

The crucial observation is that these sensitivities are successive moments of maturity. For a portfolio holding amount x_i of a bond with maturity τ_i ,

$$\underbrace{\sum_i x_i \tau_i}_{\text{duration (1st moment)}},$$

$$\underbrace{\sum_i x_i \tau_i^2}_{\text{convexity (2nd moment)}}, \quad \dots$$

$$\underbrace{\sum_i x_i \tau_i^k}_{\text{key-rate } k \text{ (} k\text{-th moment)}}.$$

So "build a portfolio that immunizes a liability up to order K " is literally the linear system

$$\sum_i x_i \tau_i^k = s_k, \quad k = 1, \dots, K,$$

where s_k are the liability's moments. This generalizes cleanly: the de Vries–Toda [1] robust formulation replaces the monomials τ^k (where k the key-rate- k duration or the k -th moment) with arbitrary basis functions $h_i(t)$ (level/slope/curvature factors, Chebyshev polynomials) and solves a worst-case (maxmin) version. Immunization is matching a stack of generalized-duration constraints.

If the bond amounts x_i could be any real numbers, this is a tidy linear (or, with robustness, a small convex) program that can be solved classically. The problem gets combinatorially hard the moment realistic constraints enter: a bond is held or not held (cardinality), weights come in discrete lot sizes, the manager wants the smallest set of bonds that immunizes, or wants to know which few bonds, if added or dropped, would break the match. These are discrete constraint-satisfaction problems over a fixed universe of bonds. Choose the subset that satisfies as many moment-matching constraints as possible and that is exactly the kind of "satisfy as many linear constraints as you can" problem where exhaustive classical search blows up.

The moment-matching matrix has entries $H_{k,i} = \tau_i^k$, a Vandermonde / locator matrix. Vandermonde systems are not generic noise.

They are precisely the parity-check structure of Reed–Solomon and BCH error-correcting codes, where each codeword position has a "locator" α_i and the parity checks are its powers α_i^k . Under a map from maturities τ_i to distinct field elements α_i , "which bonds break the immunization?" becomes "which positions carry the error?" a classical decoding problem, solvable by the efficient Berlekamp–Massey algorithm.

That is the bridge to DQI. DQI turns "maximize satisfied linear constraints" into a quantum state-preparation problem whose only hard step is decoding the constraint matrix and it amplifies good solutions cheaply exactly when that matrix is an algebraically-decodable code. Most portfolio problems (dense QUBOs) are DQI's worst case. Immunization is the opposite: its moment structure is natively the Vandermonde/locator form DQI wants.

4 Fixed-Income Immunization via DQI

4.1 From immunization to a decoding problem

Consider a financial institution with a stream of liabilities paying $f_n > 0$ at dates $t_1 < \dots < t_N$, to be hedged with a universe of n bonds of distinct maturities $\tau_1 < \dots < \tau_n$. Under a (continuously compounded) yield curve, the interest-rate sensitivities of a portfolio holding $x_i \in \{0, 1\}$ of bond i are the successive *moments of maturity*: the duration matches the first moment, the convexity the second, and the key-rate- k duration the k -th,

$$\underbrace{\sum_i x_i \tau_i}_{\text{duration}}, \quad \underbrace{\sum_i x_i \tau_i^2}_{\text{convexity}}, \quad \dots, \quad \underbrace{\sum_i x_i \tau_i^k}_{\text{key-rate } k}. \quad (8)$$

Immunizing a liability up to order $2t$ is therefore the requirement that the portfolio reproduce the liability's moment profile $\{s_k\}$,

$$\sum_i x_i \tau_i^k = s_k, \quad k = 1, \dots, 2t. \quad (9)$$

We map the $n = 2^m - 1$ maturities onto the distinct nonzero *locators* $\alpha_i = \alpha^i$ of the Galois field $\mathbb{F}_{2^m} = \text{GF}(2^m)$, where α is a primitive element. Under this map the coefficient matrix of (9) acquires entries

$$H_{k,i} = \alpha_i^k = \alpha^{ki}, \quad (10)$$

which is a Vandermonde (locator) matrix, precisely the parity-check matrix of a narrow-sense

binary BCH code of length $n = 2^m - 1$ correcting t errors. Reduced to its independent rows over $\mathbb{F}_2$, $H \in \mathbb{F}_2^{(mt) \times n}$ is the BCH parity check, with BCH-bound minimum distance $d_{\min} = 2t + 1$. The financial question "which few bonds, if included or omitted, break the immunization match?" is thus identical to *minimum-weight syndrome decoding* of this code: given a syndrome s , recover the lowest-weight error pattern y with $Hy = s$. For BCH/Reed–Solomon codes this is solved algebraically by the Berlekamp–Massey algorithm followed by a Chien search [3].

Modelling choice. Eq. (10) reinterprets the moment equations (9) over $\text{GF}(2^m)$ rather than over $\mathbb{R}$. There is no structure-preserving map $\mathbb{R} \rightarrow \text{GF}(2^m)$; this is a deliberate modelling decision. The resulting instance answers a *combinatorial* version of immunization, which small subset of bonds is the "odd-one-out" for the binarised moment syndrome, rather than the exact real-valued duration-matching problem, which a classical solver would solve as a classical linear or quadratic program. What is genuine is the algebra: the moment structure of (8), the Vandermonde matrix (10), and the reduction to algebraic decoding are the same objects a fixed-income desk uses.

We pose the combinatorial immunization instance as a max-XORSAT problem, the class solved by DQI. Setting $B^\top = H$, the constraint matrix is $B = H^\top \in \mathbb{F}_2^{m \times n}$, and the *dual code*

$$C^\perp = \{d \in \mathbb{F}_2^n : B^\top d = 0\} \quad (11)$$

is exactly the BCH code. The liability enters only through the right-hand side: its moments, reduced into the field, give a target syndrome

$$v = H^\top s_L \bmod 2 \in \mathbb{F}_2^m, \quad (12)$$

where s_L is the m -bit representation of the liability's ideal locator $L \in \text{GF}(2^m)$. This mirrors how an immunization desk operates: a fixed bond universe (fixed B) and a varying liability (varying v). The objective is the number of satisfied immunization constraints,

$$f(x) = |\{i : (Bx)_i = v_i\}|, \quad \max_{x \in \mathbb{F}_2^n} f(x), \quad (13)$$

whose optimum is the bond pattern whose locator syndrome best reproduces the liability's; a perfect combinatorial immunization corresponds to $f(x) = m$ (zero field-syndrome residual).

4.2 Quantum circuit for the decoder

The decoding step admits a reversible quantum circuit implementation whose cost depends on the correction radius t :

- $t = 1$ (**single odd-one-out bond**). The error-locator polynomial is $\sigma(x) = 1 + S_1x$ with a single root, so the m -bit syndrome is the binary address of the unique erroneous position. Uncomputing $|y\rangle$ reduces to a sequence of multi-controlled- X gates, each flipping an error qubit conditioned on $|x\rangle$ encoding its address. Routing these through a shared v -chain ancilla register (spending $n - 2$ clean ancillas restored to $|0\rangle$) trades width for depth, which is favourable on depth-limited NISQ hardware and yields a 30–41% reduction in two-qubit (CZ) gate count relative to the ancilla-free decode. This case is built and simulated end to end.
- $t \geq 2$ (**joint duration/convexity/higher-moment matching**). The same linear stages are retained, with the locator stage realised either by exact pattern enumeration (for verification) or by a Berlekamp–Massey datapath over $\text{GF}(2^m)$ (syndrome computation, key-equation solve, Chien search).

5 Problem instances and models

The construction of Sec. 4 defines a whole family of instances parametrised by the field extension m (hence the bond universe size $n = 2^m - 1$), the correction radius (distance of the code) t , and the amplification order ℓ . The results that follow are not a single experiment but a controlled sweep over this family, together with one deliberately *off-family* instance used as a negative control. We collect the instances here so the Results section can refer to them by name. Table 1 summarises them.

5.1 The algebraic BCH family (BCH-7/15/31)

The core instances are narrow-sense binary BCH codes of length $n = 2^m - 1$ over $\text{GF}(2^m)$, built exactly as in Eqs. (10)–(13). The $n = 2^m - 1$ bond maturities are mapped to the nonzero locators $\alpha_i = \alpha^i$ of the field, the moment-matching matrix becomes the Vandermonde parity check

$H_{k,i} = \alpha^{ki}$, and we set $B^\top = H$ so that the dual code is the BCH code. A liability is encoded by drawing a weight- t “odd-one-out” bond pattern y_L , forming its field syndrome $s_L = Hy_L \bmod 2$, and imprinting the target $v = Bs_L \bmod 2$; the planted optimum is then $x^* = s_L$ with $f(x^*) = m$ (zero residual). This mirrors a desk operating a fixed bond universe (fixed B) against a varying liability (varying v).

Because the constraint matrix is a BCH/Reed–Solomon parity check, the in-circuit decoder for every algebraic instance is the *Berlekamp–Massey* algorithm (realised as a reversible Berlekamp–Massey/Peterson datapath; this is what “algebraic” denotes throughout). It is precisely the existence of this structured decoder, rather than the iterative belief propagation forced on a generic matrix, that lets DQI amplify cheaply on these instances.

The three sizes play distinct roles:

- BCH-7 ($m = 3, n = 7$): the workhorse. Its full decoder circuit is 11 qubits (7 error + 3 syndrome + 1 weight); it is the instance we sweep over the three decoder regimes, run the genuine $t = 2$ decoder on, and carry all the way onto hardware.
- BCH-15 ($m = 4, n = 15$): a mid-size point (21 qubits) that confirms the algebraic advantage and gate-count trend persist beyond the smallest case, while still being classically simulable.
- BCH-31 ($m = 5, n = 31$): the hardware scale-up. Its full decoder (39 qubits) is past what we simulate, but the $t = 1$ collapse (below) brings it onto the QPU.

For each algebraic instance we additionally run two structure-blind decoders on the *same* constraint matrix as controls: Gaussian/Gauss–Jordan elimination (GJE), which carries no code structure, and (where simulable) generic belief propagation (BP). These isolate the claim that amplification comes specifically from the algebraic decoder, not from the instance.

5.2 The $t = 1$ bijection collapse (hardware form)

For $t = 1$ over $\text{GF}(2)$ the single-error location-to-syndrome map is a bijection: the error-locator

Table 1: The problem instances exercised in this work. Each algebraic instance is a narrow-sense binary BCH code of length $n = 2^m - 1$ over $\text{GF}(2^m)$ correcting t errors; “check bits” is the size mt of the binary syndrome register. “Algebraic” denotes the Berlekamp–Massey decoder (a reversible Berlekamp–Massey/Peterson datapath in-circuit). The LDPC instance is off-family: a generic sparse parity check with no algebraic decoder, decodable only by belief propagation (BP), included as a negative control.

label	code / field	bonds n	t	check bits	ℓ	decoder (regime)	role
BCH-7	BCH / $\text{GF}(2^3)$	7	1	3	1, 2, 3	Berlekamp–Massey	primary
BCH-15	BCH / $\text{GF}(2^4)$	15	1	4	1, 2	Berlekamp–Massey	mid-size
BCH-31	BCH / $\text{GF}(2^5)$	31	1	5	1	Berlekamp–Massey	hardware
BCH-7-T2	BCH / $\text{GF}(2^3)$	7	2	6	2	Berlekamp–Massey	genuine
RS-SYM	RS / $\text{GF}(2^m)$	—	1	—	—	Berlekamp–Massey + Forney	symbolic
LDPC	sparse LDPC, $d_{\min} = 3$	6–7	—	—	1	belief propagation	unstructured

polynomial $\sigma(x) = 1 + S_1x$ has its unique root at the address S_1 , so the one-hot error register and the entire reversible decode/uncompute collapse onto the bare m -bit syndrome register. The 7-bond computation reduces to preparing an m -qubit DQI amplitude state (built from the prescribed optimal weights and the v -phase signs, not hardcoded), applying $H^{\otimes m}$, and measuring. This is the form actually executed on hardware: it cuts BCH-7 from 11 qubits / 154 routed CZ to 3 qubits / 5 routed CZ, and BCH-31 to 5 qubits / ~ 29 routed CZ — both under the ~ 40 -CZ coherence wall (Sec. 7 reports both the full and collapsed runs).

5.3 The genuine multi-error instance (BCH-7-T2)

The $t = 1$ instances correct a single error; the proper DQI object is a multi-error decoder run reversibly in superposition. BCH-7-T2 is the same 7-bond code at $t = 2$ — now matching duration *and* convexity (the first two maturity moments) jointly. Here the syndrome register is $mt = 6$ bits and the decoder is a genuine reversible Berlekamp–Massey/Peterson datapath. The key simplification is that for $t = 2$ over $\text{GF}(2^m)$ the locator coefficients reduce to $\sigma_1 = S_1$ and $\sigma_2 = S_3S_1^{-1} + S_1^2$, so the whole decode needs only *two* genuine field multiplications (all squarings and inverses being $\text{GF}(2)$ -linear, hence CX-only nets) followed by a linear Chien search. This instance is the bridge from the hardware showpiece to the real, polynomially-scaling DQI decoder; it lives in simulation (28 qubits, 632 CZ).

5.4 The symbol-level Reed–Solomon instance (RS-SYM)

Binary BCH answers only *which* bond is the odd-one-out: every error magnitude is 1. A literal real-valued immunization instead carries a $\text{GF}(2^m)$ symbol per bond — the bond amount — so each error has both a location and a magnitude. RS-SYM is a $t = 1$ Reed–Solomon instance that recovers the magnitude via the Forney formula $Y = S_1^2/S_2$ by reversible field arithmetic. It exists to close the “binary reduction is load-bearing” caveat of Sec. 4: it demonstrates that the genuine, real-valued immunization is decodable in superposition at a known gate cost (37 qubits, 668 CZ).

5.5 The unstructured control (LDPC)

We include one off-family instance: a genuine sparse diversification portfolio (6-7 variables) whose constraint matrix is an unstructured LDPC parity check with dual minimum distance $d_{\min} = 3$. It carries no algebraic decoder, so the only thing that can amplify it is belief propagation. This is the negative control that draws the dividing line: real, unstructured finance problems sit in the costly BP regime, whereas immunization sits natively in the cheap algebraic regime by construction.

6 Classical decoding and validation

Step 6 of the state preparation (Sec. 2) is a unitary embedding of a *classical* algebraic decoder, and every quantum number we report rests on a classical computation underneath it. We collect

that scaffolding here, ahead of the results, because the same three objects support every entry in Sec. 7: the decoder that *defines* the correct answer, the exact bit-walk that *certifies* the circuit computes it, and the statevector reference that certifies the amplitudes are right.

6.1 The classical decoder

Section based on [3]

For an instance with parity check H (Eq. (10)), the syndrome of an error pattern y is $S = Hy$, whose components $S_j = \sum_{i: y_i=1} \alpha_i^j$ are the power sums of the locators of the errored bonds. Recovering the lowest-weight y from S — the financial “which few bonds break the immunization?” — is syndrome decoding, solved algebraically in three stages:

1. **Key equation.** Find the error-locator polynomial $\sigma(x) = \prod_e (1 - \alpha_e x)$, whose roots mark the errored positions, from the syndromes via Berlekamp–Massey (equivalently Peterson).
2. **Chien search.** Evaluate σ at every field element to read off its roots, i.e. the error locations.
3. **Forney.** For non-binary (Reed–Solomon) instances, recover each error *magnitude* Y_e from S and σ .

At the small correction radius (distance) we target the key equation collapses to a closed form, which is exactly what keeps the reversible circuit shallow:

- BCH-7/15/31 ($t = 1$): $\sigma(x) = 1 + S_1 x$ has a single root at the address S_1 , so the syndrome *is* the binary address of the unique odd-one-out bond — no field multiplication, and the decode is a pure permutation of basis states.
- BCH-7-T2 ($t = 2$): $\sigma_1 = S_1$ and $\sigma_2 = S_3 S_1^{-1} + S_1^2$, so the entire decode needs only two genuine $\text{GF}(2^m)$ multiplications — squarings and inverses are $\mathbb{F}_2$ -linear, hence CX-only nets — followed by a linear Chien search.
- RS-SYM (symbol level): the Forney magnitude is $Y = S_1^2/S_2$, one multiply and one inverse.

This classical decoder plays two roles at once. Classically it is the *reference solution* and the optimum x^* against which $P(\text{opt})$ is scored. In the circuit it is the *operation Step 6 must implement reversibly*; the two passes below certify exactly that, first as a permutation, then as an amplitude.

6.2 Validating the circuit: exact bit-walk

A reversible decoder is correct iff it realises the right permutation of computational-basis states *and* returns every ancilla to $|0\rangle$; a single ancilla left dirty silently entangles and kills the downstream interference. The decoder blocks contain no Hadamards or rotations (they are built from $X/\text{CX}/\text{CCX}/\text{MCX}$ only) so they are exact permutations, and we verify them classically with no statevector at all: we emit the circuit as a gate list and propagate each basis state through it by bit operations. For every error pattern of weight $\leq t$ we check that (i) the codeword register decodes to the locator/magnitude the classical decoder predicts and (ii) every ancilla returns to $|0\rangle$. This passes exhaustively, all 29 weight- ≤ 2 patterns for BCH-7-T2 (15 ancillas restored), all 50 single-symbol errors for RS-SYM (10 ancillas restored). Being a bit-exact statement about a permutation, the check is instant and logically prior to any amplitude simulation.

6.3 Validating the amplification: statevector reference

A correct permutation does not by itself guarantee that the *assembled* DQI circuit amplifies: the weights w_k , the v -phases, and the Fourier step must combine to the right amplitudes. We check this with full statevector simulation on LUMI CPU nodes (Qiskit Aer), comparing the simulated output distribution against the ideal DQI distribution computed directly from the semicircle weights. The agreement sits at the shot-noise floor: for BCH-7-T2 the 28-qubit circuit gives $P(\text{opt}) = 0.3713$ against ideal 0.3714 and $\langle s \rangle = 5.680$ against ideal 5.679 ($\max |\Delta| < 2 \times 10^{-4}$).

Throughout, amplification is reported by one *satisfaction statistic*: for a measured bitstring x , the fraction of the m immunization constraints it satisfies, summarised by the mean $\langle s \rangle$ and by $P(\text{opt})$, the probability of sampling a constraint-optimal x (i.e. $x = x^*$). The yardstick is always

the uniform random-guess baseline, $P = 2^{-b}$ for a b -qubit solution register. Statevector simulation is feasible only to $\lesssim 28$ qubits: this is the line in Table 2 between instances we characterise by full distribution and those we characterise by gate count alone (or, under the coherence wall, by hardware).

7 Results

We report results in two layers. First, classical statevector simulations (run on LUMI CPU nodes) establish *which* problem structure DQI can amplify and at what gate cost. Second, hardware runs on the 53-qubit VTT Q50 superconducting processor establish whether that amplification survives on real NISQ devices, and locate the fidelity wall precisely. Throughout, amplification is measured by the satisfaction statistic and scored against the uniform random-guess baseline, both defined in Sec. 6, and every reversible decoder reported here has passed the bit-walk and statevector checks of that section.

7.1 The three decoder regimes

The load-bearing claim behind the whole construction is that DQI amplifies only when the max-XORSAT constraint matrix is the parity check of an *algebraically decodable* code. We tested this directly by holding the problem instance fixed and swapping the in-circuit decoder between three regimes: a shallow algebraic (Berlekamp–Massey, $t = 1$) decoder; a generic belief-propagation (BP) decoder for unstructured sparse LDPC checks; and Gaussian elimination (GJE), which carries no code structure to exploit. Table 2 reports the sweep over code size and number of constraint repetitions ℓ .

The pattern is unambiguous. The algebraic decoder lifts $P(\text{opt})$ from the random 0.125 to 0.524 at $n = 7$ and from 0.0625 to 0.350 at $n = 15$, while costing roughly five times fewer CZ gates than the BP decoder on the same instance. GJE never leaves the random floor ($P(\text{opt}) = 0.127$ vs. 0.125): with no code structure there is nothing for the interferometer to amplify. BP does amplify ($P(\text{opt}) = 0.322$ at $n = 7$) but its gate count explodes with size ($385 \rightarrow 1889 \rightarrow 7729$ logical CZ for $n = 7, 15, 31$), placing it firmly in the simulation-only regime. As expected, in-

creasing ℓ adds depth and slightly dilutes per-shot $P(\text{opt})$ while sharpening the constraint margin.

7.2 A genuinely unstructured portfolio

To check that the algebraic advantage is not an artifact of a contrived instance, we ran a separate sweep on a genuine sparse diversification portfolio (a 6–7 variable instance whose constraint matrix is an unstructured LDPC check with dual minimum distance $d_{\min} = 3$). Here only the BP decoder can amplify, it reaches $P(\text{opt}) = 0.238$ –0.260 against a random baseline of 0.0156 (a 15–17 $\times$ lift), but at ~ 1748 –2032 logical CZ on 24–25 qubits. GJE on the same instance stays flat ($P(\text{opt}) \approx 0.092$). This confirms the dividing line: real, unstructured finance problems live in the expensive BP regime, whereas immunization, because bond sensitivities are power sums $\sum_i x_i \tau_i^k$, i.e. a Vandermonde / Reed–Solomon parity check, lives natively in the cheap algebraic regime.

7.3 The genuine multi-error decoder

The $t = 1$ instances above correct a single error and amplify with $\ell/m = 1/7$. The proper DQI object is the multi-error decoder run reversibly in superposition. For BCH-7-T2 we built and ran the genuine $t = 2$ reversible Berlekamp, Massey/Peterson datapath, the two-multiply locator reduction of Sec. 6, the smallest instance in which the decoder, rather than a bijection, is the whole circuit.

Both validation checks of Sec. 6 pass for this decoder: the exact bit-walk certifies the permutation (all 29 weight- ≤ 2 patterns decode, 15 ancillas restored), and the statevector reference certifies the amplitudes ($P(\text{opt}) = 0.3713$ vs. ideal 0.3714, $\langle s \rangle = 5.680/7$ vs. ideal 5.679, on a LUMI job). On cost (Table 3) the algebraic BM datapath is 3.4 $\times$ cheaper and 5.5 $\times$ shallower than naive pattern enumeration, and crucially scales *polynomially* (two multiplies) rather than combinatorially. In the proper metric the lift is stronger at $t = 2$: $\ell/m = 2/7$ gives a 23.8 $\times$ $P(\text{opt})$ lift over random, versus 6.65 $\times$ at $t = 1$.

The symbol-level Reed–Solomon decoder (RS-SYM) carries this to the real-valued instance: it recovers the error magnitude via the Forney formula $Y = S_1^2/S_2$ in reversible field arithmetic (37 qubits, 668 CZ), bit-walk-exact on all 50 single-

Table 2: Decoder-regime sweep (LUMI statevector simulation). For each BCH code of length $n = 2^m - 1$ (= number of bonds) over $\text{GF}(2^m)$ we report logical and routed two-qubit (CZ) gate count, the mean number of constraints satisfied $\langle s \rangle$, and $P(\text{opt})$, against the random baseline. The algebraic decoder amplifies at a small fraction of the BP cost; GJE stays at the random floor; BP amplifies but is far too deep to simulate beyond $n = 7$.

n (bonds)	decoder	ℓ	qubits	logical CZ	routed CZ	$\langle s \rangle$	$P(\text{opt})$	random P
7	algebraic	1	11	86	158	5.04	0.524	0.125
7	algebraic	2	11	132	250	4.57	0.467	0.125
7	algebraic	3	11	171	324	4.32	0.401	0.125
7	BP	1	21	385	766	4.29	0.322	0.125
7	GJE	1	10	30	53	3.50	0.127	0.125
15	algebraic	1	21	264	501	9.95	0.350	0.0625
15	algebraic	2	21	382	740	9.24	0.305	0.0625
15	GJE	1	19	81	183	7.48	0.061	0.0625
15	BP	1	40	1889	3893	not simulated (> 28 qubits)		
31	algebraic	1	39	703	1427	not simulated (> 28 qubits)		
31	GJE	1	36	200	554	not simulated (> 28 qubits)		
31	BP	1	73	7729	15539	not simulated (> 28 qubits)		

Table 3: Genuine $t = 2$ decoder: reversible Berlekamp–Massey vs. pattern enumeration on the 7-bond instance.

decoder	qubits	CZ	depth
Berlekamp–Massey ($t = 2$)	28	632	749
pattern enumeration	13	2128	4156

symbol errors (Sec. 6). This closes the “binary-reduction is load-bearing” caveat: the literal real-valued immunization is decodable in superposition, at known gate cost.

7.4 Hardware: VTT Q50

Calibration. A one-CZ Bell-state smoke test on Q50 returned 97.2% in-subspace population, fixing the effective per-CZ error at $\approx 0.9\%$. This sets the coherence budget: an interference-based algorithm of N CZ gates retains roughly $(0.991)^N$ of its coherent amplitude before idle/readout loss, so the practical wall sits near $N \approx 40$ CZ ($\sim 76\%$ survival).

The full circuit decoheres (null). We first ran the complete 7-bond algebraic immunization circuit (11 qubits, 86 logical / **154 routed CZ**, depth 226; 2000 shots, bare and with FiQCI-EMS level-1 mitigation). The result is a clean null:

The solution register came out flat across all eight patterns (10.8–14.1%, all $\approx$ the uniform 12.5%) — full decoherence, not a measurement

Table 4: 7-bond full circuit on Q50 (154 routed CZ): the 3-bit solution register decoheres to uniform.

	$\langle s \rangle$	$P(\text{opt})$
random	3.50	0.125
simulator	6.34	0.830
Q50 (bare)	–	0.122
Q50 + EMS	–	0.136

bug (the readout pipeline reproduced the simulator’s $P(\text{opt}) = 0.830$ in self-check). This quantifies exactly where the NISQ wall sits for a delicate phase-interference algorithm: $(0.991)^{154} \approx 25\%$ coherent survival is already too little, and amplification — being a second-order interference effect — is the first thing to die.

Crossing the wall: the $t = 1$ bijection collapse. For $t = 1$ over $\text{GF}(2)$ the single-error location $\rightarrow$ syndrome map is a *bijection*, so the one-hot error register and the entire reversible decode/uncompute collapse into the bare syndrome register. The 7-bond computation reduces to preparing a 3-qubit DQI amplitude state (built from the prescribed optimal weights and v -phase signs — not hardcoded), applying $H^{\otimes 3}$, and measuring. This cuts **11 qubits / 154 routed CZ down to 3 qubits / 5 routed CZ**, comfortably under the coherence wall. On Q50 (2000 shots, device qubits 14/22/23, depth 12):

Table 5: Collapsed 7-bond circuit on Q50 (5 routed CZ): DQI amplifies on hardware.

	$\langle s \rangle$	$P(\text{opt})$
random	3.50	0.125
simulator	6.32	0.831
Q50 (bare)	6.02	0.756
Q50 + EMS	–	0.692

This is the headline positive result: $P(\text{opt}) = 0.756$, a $6.0\times$ **lift over random**, reaching 91% of the noiseless simulator value. The correct portfolio (the bond-3 “odd-one-out” solution) dominated the distribution by $\sim 11\times$. Error mitigation was marginally worse here (0.692), so the bare run is the primary result.

Scale-up: 31 bonds. The collapse generalizes to $\text{GF}(2^b)$ on b qubits, with state-preparation CZ growing roughly linearly in the number of bonds. The 31-bond $\text{GF}(2^5)$ instance (5 qubits, 26 logical CZ) transpiled to **29 routed CZ** via a 16-seed mapper sweep, which auto-selected a compact device patch (qubits 39/45/38/31/40), beating a 34-CZ ring estimate and staying under the 40-CZ wall. On Q50 (2000 shots):

Table 6: 31-bond $\text{GF}(2^5)$ circuit on Q50 (29 routed CZ).

	$\langle s \rangle$	$P(\text{opt})$
random	17.50	0.031
simulator	25.78	0.674
Q50 (bare)	17.46	0.154
Q50 + EMS	–	0.139

The optimum x^* was the single dominant string (307/2000 shots), giving $P(\text{opt}) = 0.154$, a $4.9\times$ **lift over random**. The distribution is bimodal on $\{15, 31\}$: every non-optimal string satisfies exactly 15/31 constraints, a signature of the $t = 1$ bijection structure rather than of noise. The lift is honestly milder than at 7 bonds, both because the run sits nearer the fidelity wall (29 vs. 5 CZ) and because the relative constraint margin ℓ/m shrinks from 1.81 to 1.66 with size.

7.5 Summary

Taken together: in simulation, DQI amplifies cheaply and only when the constraint ma-

trix carries algebraic code structure, exactly the structure fixed-income immunization possesses by construction, and the genuine multi-error Berlekamp–Massey decoder amplifies more strongly still ($23.8\times$) at polynomial cost. On hardware, the full decoder is past today’s ~ 40 -CZ coherence wall and decoheres to uniform at 154 CZ, but the $t = 1$ bijection collapse brings the circuit under the wall and produces genuine above-random amplification on the VTT Q50 at two problem sizes ($6.0\times$ at 7 bonds, $4.9\times$ at 31 bonds). The remaining gap to running the genuine decoder on hardware is quantified: survival ≥ 0.5 over the 632-CZ $t = 2$ circuit requires a per-CZ error of $\lesssim 0.11\%$, roughly one to two superconducting generations beyond Q50’s current 0.9%.

8 Discussion

Our results support a narrow but, we believe, well-defined claim: among finance problems, fixed-income immunization is unusual in that its native constraint structure, the Vandermonde form of moment matching, is precisely the algebraically decodable structure that lets DQI amplify at low circuit cost.

8.1 What was actually solved

To avoid overclaiming, it is worth restating plainly what the experiments established. We did *not* optimize a real bond book. We solved a finite-field, combinatorial surrogate, “which weight- t subset of bonds is the odd-one-out for the binarised moment syndrome over $\text{GF}(2^m)$?”, which is a faithful representation of immunization’s *algebra* (its moment matrix, its locator structure, its reduction to syndrome decoding) but not of its real-valued cash flows. Within that surrogate, the simulation evidence is solid: the genuine multi-error Berlekamp–Massey decoder amplifies $P(\text{opt})$ $23.8\times$ over random at polynomial gate cost, and the symbol-level Reed–Solomon instance shows the magnitude-recovery step (Forney) is likewise realisable in superposition.

8.2 Limitations

- **Real-valued cash flows are not preserved.** The moment equations (9) are reinterpreted over $\text{GF}(2^m)$ rather than over $\mathbb{R}$. There is

no structure-preserving map $\mathbb{R} \rightarrow \text{GF}(2^m)$, so the instance answers a combinatorial question, not the exact duration/convexity-matching program a desk solves.

- **The $t = 1$ hardware collapse is not the full DQI decoder.** The circuits run on Q50 exploit the single-error location $\rightarrow$ syndrome bijection, which collapses the decode to an m -qubit amplitude state. This demonstrates hardware amplification but bypasses the genuine reversible decoder.
- **Current hardware cannot run the genuine $t \geq 2$ decoder.** The full algebraic decoder decoheres to uniform at 154 routed CZ on Q50.
- **The advantage is structural, not broad finance optimization.** The result identifies one finance-motivated structure that fits DQI; it does not imply DQI is advantageous across portfolio optimization, most of which (dense QUBOs) sits in DQI’s expensive regime.

9 Conclusions

DQI converts max-LINSAT optimization into a decoding problem, and amplifies cheaply only when the constraint matrix is the parity check of an algebraically decodable code. We identified fixed-income immunization as a finance-motivated source of exactly this structure: bond sensitivities are maturity moments, so moment matching is a Vandermonde/locator system, i.e. a BCH/Reed–Solomon parity check. Posing a finite-field combinatorial surrogate of immunization as max-XORSAT, statevector simulations confirmed that amplification is tied specifically to the algebraic decoder, and hardware runs on the VTT Q50 processor produced above-random amplification ($6.0\times$ at 7 bonds, $4.9\times$ at 31 bonds) once the $t = 1$ collapse brought the circuit under the coherence wall. These results do not constitute a production fixed-income optimizer; they establish immunization-style moment matching as a concrete, finance-motivated structure in which DQI’s decoding requirement is naturally satisfied, and they quantify the hardware gap remaining before the genuine multi-error decoder can be run.

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